ITU Noticeboard Patrol: Autonomous Observation and Anomaly Detection

YZV406E Robotics - Project Proposal (Fall 2025)

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1 Project Topic

Design and simulate a mobile robot that patrols university corridors, approaches noticeboards, captures high-resolution poster images, and performs intelligent information extraction. The system utilizes Cloud-based Vision Language Models (VLM) to detect expired, outdated, or duplicated announcements and generates structured reports using a persistent memory system.

2 Problem Definition & Motivation

Highly used buildings (e.g., Architecture, EEF) contain many noticeboards whose content changes frequently. Manually checking for expired or duplicate announcements is tedious. We propose an autonomous system that safely navigates crowded corridors, aligns to boards, and leverages the Google Gemini API to analyze poster content. This approach solves the problem of detecting complex anomalies (e.g., past-due dates, duplicate flyers) while optimizing onboard computational resources.

3 Environment & Platform

- **Simulation:** Gazebo Harmonic indoor corridor environment populated with multiple noticeboards, **benches**, and dynamic obstacles (moving pedestrians). Real-world poster images will be used as textures for realistic validation.
- **Robot Platform:** TurtleBot-class differential drive robot equipped with a 2D LIDAR for mapping and an RGB camera for board detection.
- **Software Stack:** ROS2 Humble.
- **Computational Strategy:** To manage limited VRAM, heavy visual reasoning is offloaded to the Cloud (Gemini API), simulating a modern edge-cloud architecture.

4 High-Level Requirements

1. **Safe Navigation & Approach:** Build a 2D map, localize, avoid static (benches) and dynamic (pedestrians) obstacles, and perform fine-grained docking to a board viewing pose.

2. **Perception & Cloud Integration:** Identify noticeboards and transmit images to the Google Gemini API. The system must extract relevant content (Titles, Dates, Summaries) and detect anomalies such as **expired dates** or **duplicated posts**.
3. **Memory & Data Logging:** Maintain a database recording Board IDs, timestamps, captured images, extracted fields, and anomaly flags for reporting and traceability.
4. **Revisiting Logic (Planning):** A high-level planner that schedules visits to boards previously flagged as "Expired" or "Unclear" to verify their status.
5. **Testing & Metrics:** Evaluate navigation safety, approach accuracy, and information analysis precision/recall (demonstrated via video).

5 Method & Approach

5.1 Perception (Cloud-Based Information Extraction)

- **Observation Capture:** The robot identifies noticeboards using onboard detection, aligns its view to ensure sufficient framing and resolution, and captures the image.
- **VLM Analysis (Gemini API):** Instead of error-prone local OCR, the captured image is sent to the Gemini API. The prompt requests a structured JSON output: { `"title": string, "event_date": date, "is_expired": boolean, "is_duplicate": boolean` }.
- **Advantage:** This hybrid method allows for understanding context (e.g., "Next Monday") and detecting duplicates by comparing semantic summaries in the cloud.

5.2 Localization, Navigation, and Control

- **Mapping Localization:** The robot constructs an internal map of corridors and updates its position continuously to navigate accurately among benches and pedestrians.
- **Fine Positioning:** Upon reaching a target, the robot switches to a visual servoing behavior to stop at a suitable viewing distance (approx. 50cm) and face the board frontally.

5.3 Memory System and High-Level Planning

- **Structured Routine:** The full behavior follows the sequence:

  $$Navigate \rightarrow AlignView \rightarrow Capture\&Analyze(Cloud) \rightarrow LogtoMemory \rightarrow Decide(Next/Revisit)$$

- **Memory Structure:** A persistent store records the state of each board (e.g., *Board_ID: 3, Status: Expired, Last_Seen: 14:00*).
- **Revisiting Logic:** If the API flags an item as expired, the planner marks it for a "Reinspection" tour to confirm if facility staff have removed it.

6 Individual Responsibilities

Member	Primary Task	Sample Subtasks
Metin Ertekin Küçük	Perception & Cloud API	Implementing Board Detection; Integrating Google Gemini API for extraction (Title, Date, Duplicates); JSON parsing; Video duties.
Abdullah Saatçi	Navigation	Developing safe avoidance (Pedestrians/Benches); implementing fine-positioning/docking behavior; Presentation duties.
Emre Saraç	Planning & Memory	Designing the Memory System (Logging IDs, timestamps); implementing the revisiting logic; coordinating system integration.

7 Timeline

Milestone	**Dates**	**Details**
Project Start & Setup	5-12 Nov 2025	Repository setup, Gazebo environment with benches and real posters.
Mapping & Navigation	13-19 Nov 2025	Map creation; Basic navigation; Initial implementation of docking.
Perception Integration	20-26 Nov 2025	Board detection; Gemini API connectivity tests; Prompt engineering.
Memory & Logic	27 Nov - 3 Dec 2025	Connecting API outputs to Memory; Implementing re-inspection logic.
Testing	4-10 Dec 2025	Full system loop test: Explore $\rightarrow$ API Call $\rightarrow$ Memorize $\rightarrow$ Revisit.
Polish & Docs	11-17 Dec 2025	Final adjustments, report writing, demo preparation.
Final Demo	18-29 Dec 2025	Full scenario presentation.